

Controlled-error bound on the enumerated-reading selection margins

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

B1 asserts $\Delta F[\mathcal{R}] = F[\mathcal{R}] - F[\mathcal{R}_H] > 0$ for every enumerated reading $\mathcal{R}$ (single-shell LAM/HEX/FCC/BCC) at $\mu^2 = 0.005$, at ESTIMATOR grade – point estimates from the fast quadrature pipeline with no certified error. The ESTIMATOR-UPGRADE gate asks for a CONTROLLED error bound. This note supplies it for the single-shell readings against the dominant numerical knob (the loop-integral quadrature). Scope: single-shell ensemble; two-shell + the δI /grid knobs are registered as the same-method follow-up.

2. What "controlled error" means here

$\Delta F_{\mathcal{R}}(A)$ is even in the condensate amplitude A with $\Delta F_{\mathcal{R}}(0) = 0$ exactly ($A = 0$ is the Reading-H reference). The selection holds iff, for every reading, (a) $A = 0$ is a STRICT local minimum – the curvature

$$\kappa_{\mathcal{R}} := \Delta F''_{\mathcal{R}}(0) = \frac{2 \Delta F_{\mathcal{R}}(h)}{h^2} + O(h^2) > 0,$$

and (b) there is no DEEPER minimum at $A > 0$ (the amplitude scan's global best is at $A = 0$: no condensate). A reading with $\kappa_{\mathcal{R}} < 0$ or an interior $\Delta F < 0$ minimum would FALSIFY the selection. $\kappa_{\mathcal{R}}$ is grid-independent (unlike a " $\min_{A>0} \Delta F$ ", which scales with the amplitude-grid spacing), so it is the correct controlled-error margin.

3. Method and result

Each verdict is computed at the production fast pipeline (M_{fast} , grid at $N_{\text{pt}} = 6000$) and at a HIGH-resolution pipeline (M by direct quadrature at $N_{\text{pt}} = 20000$, installed by monkeypatching Math424's M_{fast} so its OWN certified `dF_reading/scan_reading` are reused – no formula re-transcription). $h = 0.005$ with a half-step $h/2$ quadratic-regime check.

reading	κ (fast)	κ (hires)	envelope	certified κ –env
LAM	0.85097	0.85098	1.7×10^{-5}	0.85095
HEX	2.55417	2.55422	5.4×10^{-5}	2.55411
FCC	3.40777	3.40784	7.5×10^{-5}	3.40769
BCC	5.11625	5.11637	1.2×10^{-4}	5.11613

Every reading has $\kappa_{\mathcal{R}} > 0$ with the M-quadrature envelope $< 0.1\%$ of κ (five orders below κ); the binding (smallest) reading is LAM, $\kappa = 0.851 \pm 1.7 \times 10^{-5}$. Both pipelines find every reading

collapses to $A = 0$ (no condensate: the global ΔF minimum is at $A = 0$). The $\Delta F(0) = 0$ identity holds exactly in both pipelines. Thus the single-shell selection margins are CONTROLLED-ERROR positive: $A = 0$ is a strict minimum for every reading and the verdict is not a fast-quadrature artefact.

4. What this advances and what remains

ADVANCED: the single-shell enumerated $\Delta F > 0$ verdict is now controlled-error (curvature-certified, no condensate, M-quadrature envelope $< 0.1\%$). REMAINING for full ESTIMATOR-UPGRADE closure: (i) the two-shell ensemble (same curvature + no-condensate method); (ii) the δI quadrature and amplitude-grid knobs (here held at fixed resolution in both pipelines; the M knob is dominant because M enters ΔF several times and drives the gap solve). These are registered follow-ups; the gate stays OPEN pending them and operator sign-off.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, 6 code-discipline)

- (α) “Only the M-quadrature was varied.” VALID and scoped: M is the dominant knob (multiple appearances + drives the gap solve, which also uses M_{fast} and is therefore refined here too). δI and the amplitude grid are named as follow-ups; the curvature is grid-independent, and the no-condensate scan uses the same 121-point grid in both pipelines so a condensate would have to hide from BOTH.
- (β) “ $\kappa = 2\Delta F(h)/h^2$ may be A^4 -contaminated.” DISMISSED: the h vs $h/2$ check agrees to $< 2\%$ for every reading at $h = 0.005$, confirming the finite difference measures $\Delta F''(0)$, not the quartic.
- (γ) “No formula re-transcription – sure?” The hires pipeline reuses Math424’s OWN `dF_reading/scan_reading` via a one-line monkeypatch of M_{fast} ; the ΔF algebra is never copied, eliminating transcription risk (code-discipline rule 1).

Quantitative sanity check (mandatory): magnitude – envelopes $\sim 10^{-5}$ vs $\kappa \sim 1\text{--}5$ ($< 0.1\%$); limit-case – $\Delta F(0) = 0$ exactly in both pipelines; sign-direction – $\kappa > 0$ means $A = 0$ is a strict minimum, the selection-preserving direction; consistency – BCC has the largest curvature yet its A^4 contamination forced the smaller h , consistent with a steeper landscape.

6. Result footer

Result ID:	B1-RH-ENUM / ESTIMATOR-UPGRADE (enumerated single-shell)
Precise statement:	Controlled-error: for LAM/HEX/FCC/BCC at $\mu^2=0.005$, $A=0$ is a STRICT minimum ($\kappa_R = dF_R''(0) \geq 0.85$) and no reading condenses, at BOTH $N_{\text{PT}}=6000$ and 20000 ; M-quadrature envelope $< 0.1\%$ of κ (binding LAM $\kappa=0.851 \pm 1.7e-5$). The estimator-grade $dF>0$ is upgraded to a controlled-error bound for the single-shell ensemble.
Scope:	Single-shell enumerated readings; M-quadrature knob. Two-shell + dI /grid knobs = same-method follow-up.
Dependencies:	Math424_AddA_reading_uniqueness (<code>dF_reading</code> , <code>scan_reading</code> , <code>M_of</code>); production constants.
Evidence grade:	EXECUTED (<code>estimator_upgrade_enumerated.py 7/7</code>) with a

Reproduction command:	two-resolution quadrature envelope. python codes/vacuum/estimator_upgrade_enumerated.py
Expected output:	7/7 PASS; kappa table; no-condensate at both resolutions; binding LAM kappa=0.851.
Falsification gate:	A reading with kappa <= 0 or an interior dF<0 minimum at either resolution (would refute the selection); a two-shell reading that condenses.
Tier before / after:	No tier/gate flip. ESTIMATOR-UPGRADE: single-shell controlled-error ADVANCE (M-quadrature); gate stays OPEN pending two-shell + dI knobs + operator sign-off.
No-overclaim statement:	Controlled error w.r.t. the M-quadrature only, single- shell only. NOT a full ESTIMATOR-UPGRADE closure. The B1 T6 selection SIGN was already established; this upgrades the enumerated MARGINS' error grade.
Next required action:	Two-shell ensemble + dI/amplitude-grid envelopes by the same method; then ESTIMATOR-UPGRADE closure is an operator decision.